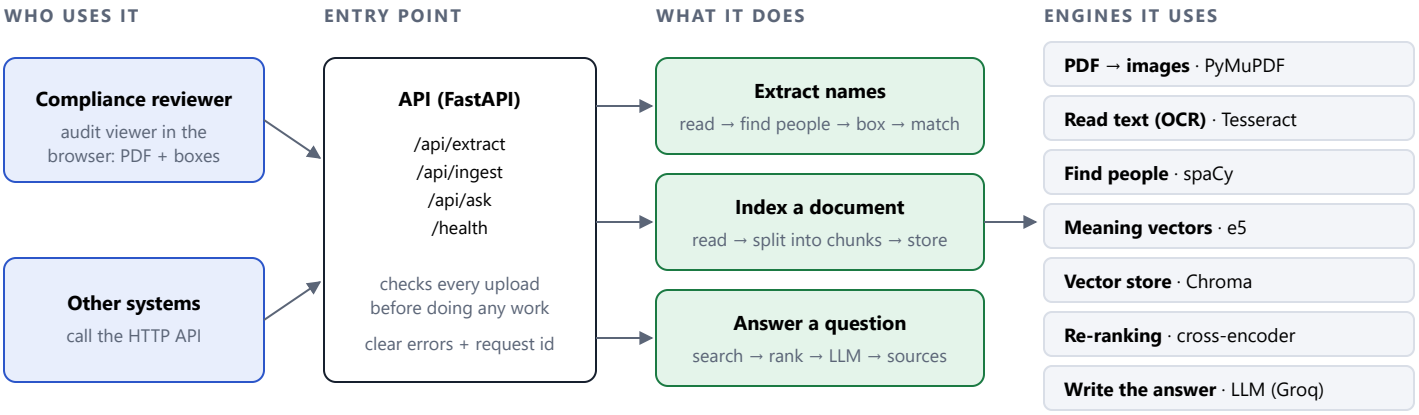
Document Intelligence Service — Design

Give it a scanned PDF and it returns every person mentioned, with the page and the exact box where the name sits, so a reviewer can check it. It also compares those names with a list you send, and answers questions about uploaded documents, citing the page each answer comes from.

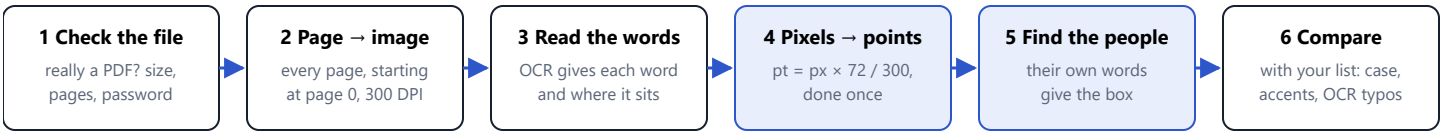
100 % / 98 %	**54 of 54**	**15 of 15**	**0**
of names found are real people / of real people are found	boxes checked by re-reading what is inside them	questions answered correctly, three runs in a row	made-up sources in any answer

1 · How the pieces fit together



Each engine sits behind its own small interface: it can be swapped through configuration, or replaced by a fake in tests.

2 · Finding names and where they are

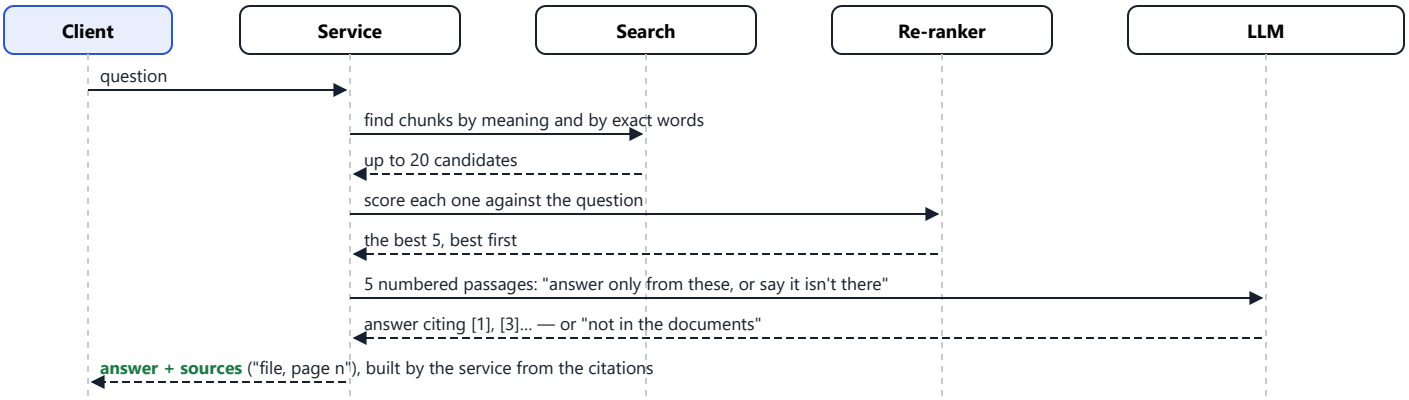


Real result on company_memo.pdf

"Robert Chen" → page 0, x 59.76, y 266.88, width 53.04, height 7.2 (PDF points, top-left origin) → matches "Robert Chen", score 1.0

- **Boxes are in PDF points, not image pixels** (72 per inch, origin at the top-left), so they line up with the PDF whatever resolution was used to read it.
- **The box comes from the person's own words.** The name finder marks a person in the page text; we take exactly the OCR words it covers and draw the box around them. A name that appears five times gets five boxes, and titles like "Dr." stay outside.
- **Matching compares whole names**, ignoring case, accents and punctuation: *María González* matches *Maria Gonzalez*, but *Ana* does not match *Ana María Rodríguez*.

3 · Answering a question



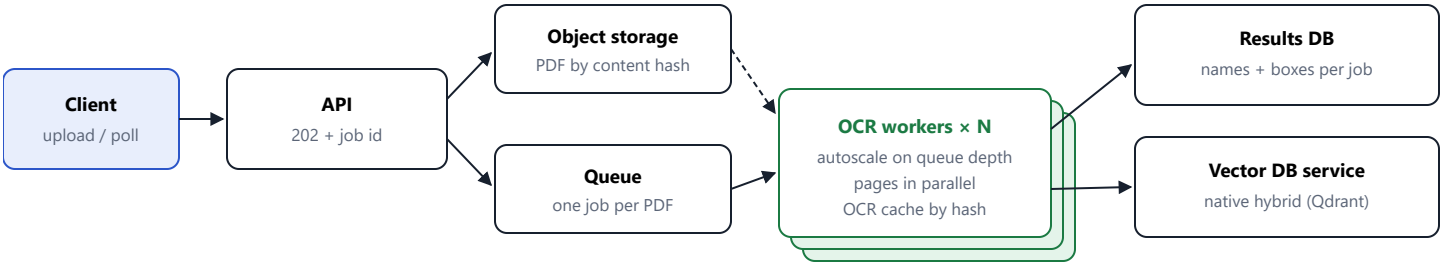
- **Indexing.** Each uploaded document is read the same way, then split into chunks of whole paragraphs (never cutting a word, never crossing a page). Chunk ids start with a fingerprint of the file, so a second document can't overwrite the first and uploading the same file twice doesn't duplicate it.
- **Two searches, then a ranking.** A search by meaning finds paraphrases; a search by exact words finds names and figures the first one misses. A re-ranker then reads each candidate next to the question and keeps the best five.
- **The LLM decides if the answer is there.** It answers only from those five passages, or says it isn't in the documents. We first tried a score cut-off to decide that, and dropped it: it rejected easy questions like "Who is the CEO?".
- **Sources can't be invented.** The LLM only cites passage numbers; the service turns them into "file, page n" from the passages it actually sent.

4 · Technology choices and trade-offs

NEED	CHOICE	WHY	TRADE-OFF
Read scanned text	Tesseract	Free, runs on CPU, gives each word's position	Struggles with poor scans; a cloud OCR could replace it
PDF → images	PyMuPDF	Fast, knows page sizes in points	AGPL licence; pypdfium2 if that matters
Find people	spaCy (large English model)	Measured best on our samples; the small model mistook a project and a magazine for people	~400 MB more; weak on ALL-CAPS text
Compare names	rapidfuzz	Whole-name similarity, fast	A fixed threshold (0.90): configurable
Search by meaning	e5 (multilingual) + Chroma	Spanish and English, no extra servers	One process writes; a database service at scale
Rank results	Multilingual cross-encoder	~1.5 s per question on CPU	A stronger model took 21 s on CPU
Write answers	Groq, gpt-oss-120b	Free and fast; any OpenAI-compatible provider by config	Rate limits on the free tier; text leaves the machine

5 · Scaling to 1000+ PDFs an hour

1000 PDFs an hour is one every 3.6 seconds. Reading a page takes 2–3 seconds of CPU, so with 5 pages per PDF that is about 4 busy cores (8 with headroom). The heavy work moves out of the request: the upload returns a job id right away, and workers that scale with the queue do the reading.



Also: GPU or hosted reranker/embedder · paid LLM tier with concurrency sized to its rate limit · OpenTelemetry traces per stage · lower DPI (200) as a speed/accuracy knob.

6 · What can go wrong and what happens

FAILURE	WHAT WE DO	CALLER SEES
Not a PDF, empty, too big, too many pages, password	Checked before any work starts	4xx with a specific code
OCR engine missing or stuck	Time limit per page	503
Poor scan	Words the OCR is unsure of are dropped, so no box is drawn over noise	fewer names
LLM slow, busy or down	Time limit and automatic retries; names and indexing keep working	503 / 502
Answer not in the documents	The LLM says so; an answer that cites nothing counts as "not found"	"not found"
A document tries to give the LLM orders	Document text is marked as data; the LLM has no tools	normal answer
Same file uploaded twice	Same fingerprint, so it replaces itself	no duplicates
Text in ALL CAPS	Known limit of the name finder	documented

7 · Choosing a model by question difficulty

Simple lookups ("Who is the CEO?") go to a small, cheap model; questions that compare, add up or ask "why" across several passages go to the large one. The choice is made by plain rules after the search (how many passages look relevant, words like "compare" or "why"), not by another LLM. If the small model gives an answer without citations while the search found good evidence, the question is retried once on the large model. The rules are tuned on our test questions, for accuracy and cost.

Details: DECISIONS.md (decisions, limitations, use of AI) · run with **docker compose up --build**, then open http://localhost:8000/viewer